# Main (Written) Assignment – Cold-Chain Transport Condition Logger

*Embedded Processors UESTC & HN 2004 School of Engineering, University of Glasgow*

## *UESTC 2004 Main (Written) Assignment Assessment Criteria*

Each individual Assignment will be evaluated using the below criteria:

- Demonstrating a functional embedded system project – 30%

  This will be assessed based on the video demonstration and your design implementation (software and hardware).

- Technical report – 70%

Please adhere to coding good practice

- Meaningful variable names – please avoid names like “x”, “y”, or personal names
- Using block comments to explain what your code does at the beginning and using line comments to clarify what this line of code does.
- Consistent indentation – Indentation helps in making the code readable and easy to interpret. See an example of what to do and what to avoid, below:

**Good practice ✓**

```
#include “mbed.h”
char switch_word ;                     //the word we will send char
recd_val;                              //the received value
int main() {
   async_port.baud(9600);                          //set baud rate to 9600
recd_val = 0;
         while (1)
         {
             switch_word=0xa0;
         }
}
```

**Avoid this ✗**

```
#include “mbed.h”
char switch_word ;                    //the word we will send char
recd_val;                    //the received value
int main() {
async_port.baud(9600);                    //set baud rate to 9600
recd_val = 0;
while (1)
{
switch_word=0xa0;
}
}
```

All the best of luck.

The Embedded Processors Team,

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## *Brief:*

The aim of this project is to design and develop a **low-power embedded system for a Cold-Chain Transport Condition Logger application** capable of monitoring **temperature exposure and handling conditions** during transportation, using the **NUCLEO-L432KC mbed microcontroller development board**, the **TMP102 digital temperature sensor**, the **ADXL345 accelerometer**, and other input/output peripherals that you may find necessary (e.g. push buttons, LEDs, buzzer, display, etc. - these are your design choices).

Cold-chain logistics are widely used in food, pharmaceutical, and laboratory contexts, where products must be kept within specific environmental conditions throughout transport. By integrating environmental sensing, motion detection, and power-aware operation, the system is intended to record and report whether a transported item has been exposed to **unacceptable temperature ranges** or **excessive handling events**, while operating for extended periods on limited power.

This project places strong emphasis on **applying the embedded systems development life cycle**, from analysing a problem statement and extracting requirements, through specification, design, implementation, testing, and reflection.

## *The Assignment:*

The aim of this project is to **design, implement, and test** an embedded system that operates as a **Cold-Chain Transport Condition Logger**, using input/output peripherals connected to the **NUCLEO-L432KC** board.

You will work in **groups of two (Working individually or in a group of 3 is acceptable only if approved, in writing, by the Course Coordinator)**. By now you should have already submitted your group members on Moodle. The project must be developed collaboratively, with **clear task allocation and individual responsibility**, and evidence of how each group member contributed to the system's development.

### *Detailed Description:*

The Cold-Chain Transport Condition Logger is to be designed to monitor:

- Environmental temperature during transport, using the TMP102 sensor.
- Handling and motion events, such as vibration, shocks, or prolonged movement, using the ADXL345 accelerometer.

The system should be capable of identifying and recording events of interest, such as:

- Temperature exceeding defined upper or lower limits.
- Sudden or sustained motion indicative of rough handling.
- Transitions between operating states (e.g. idle, transport, inspection).

The method used to define:

- acceptable temperature ranges,
- motion thresholds,
- sampling rates,
- event classification logic
- and record, store, summarise, and present these events

is your design decision and must be clearly justified in your report with respect to power consumption, memory usage, and system requirements.

The system does not need to provide continuous real-time alerts during transport. Instead, it should support low-power operation and provide a clear indication or report of the transport conditions when inspected (e.g. via LEDs, buzzer, display, or serial output).

### *Hardware:*

Required:

- Breadboard
- One Nucleo L432KC mbed board and USB programming cable
- One TMP102 sensor
- One ADXL345 accelerometer

One or more of the following (based on your design choice):

- Piezo buzzer
- LEDs
- LCD
- Push buttons
- On/Off switch
- Other components available in the lab that you justify as necessary.

### *Software:*

mbed Desktop IDE.

## *Group Work Requirements & General Guidance:*

This project is assessed as a **group** design and development activity. However, you must:

- allocate **specific responsibilities** to each group member (e.g. sensor interfacing, power management, system logic, testing, documentation).
- submit a task allocation and contribution statement as part of your report, clearly detailing:
  - who was responsible for what,
  - how work was coordinated,
  - and how integration was managed.

  Failure to clearly demonstrate individual contributions may affect your overall mark.

You should also aim to:

- take a modular approach to this design problem and breakdown your system into smaller components.
- demonstrate how individual modules were designed, implemented, and tested independently before system integration.
- study the hardware and software requirements of each component and start to implement them, one at a time, e.g., start by interfacing the TMP102 and then the ADXL345, independently, or vice versa, then work on the logic that will fulfill the system requirements.

## *Design Guidance:*

Please note that this is a **design problem**, not just a development exercise.

You must:

- Apply the embedded systems life cycle, explicitly showing:
  1. Requirements analysis
  2. System specification
  3. Hardware and software design
  4. Implementation
  5. Testing and validation
  6. Reflection and improvement
- Take design decisions on:
  - temperature thresholds,
  - motion/event detection logic,

  - operating modes,
  - power saving strategies.
- Clearly state and justify:
  - all assumptions made,
  - all design choices.
- Use:
  - system architecture/block diagrams,
  - flowcharts,
  - modular software structure.
- Conduct unit testing of individual components before full system integration.

## *Technical Report:*

A formal written report is required that details the analysis, design and testing carried out for this project, is required.

Your report must cover and include the following:

1. A front page (***including your group number, group member names and GUIDs***)
2. A contents page showing the page numbers, list of tables, list of figures
3. **Introduction**: This section provides an overview of the project, including its purpose, objectives, and scope. It should also include a brief discussion of the background, with some literature references, and motivation for the project. **(Weight – 5 marks out of 70)**
4. **Methodology**: This section describes the methods used to design, implement, and test the embedded system. It should include a detailed description of the system development process including requirements analysis, justified hardware/software design, modular implementation, power-aware decision choices, testing strategy and any experimental procedures used to validate the system's functionality. **(Weight – 30 marks out of 70)**
5. **Results and Discussions**: This section presents the results of the project, including any data or observations gathered during testing. It should also include a discussion of the significance of the results and how they relate to the project's objectives. **(Weight – 20 marks out of 70)**
6. **Conclusions and Lessons Learnt**: This section summarises the key findings of the project and discusses their implications and significance for the field of embedded systems engineering. It should also highlight the project's contributions and potential impact, discuss any limitations and suggest areas for future development. **(Weight – 10 marks out of 70)**
7. **References**: This section lists all the sources cited in the report, using a consistent citation

style (IEEE style). **(Weight – 5 marks out of 70)**

8. **Appendices**: This section includes any additional information that is relevant to the project but not included in the main text of the report, such as detailed technical specifications or codes.

## *Deliverables:*

You are required to submit the following on Moodle on a single compressed file (.zip or .rar or a suitable format), that includes the following, by the ***08th of June 2026*****:**

1. A formal written report as highlighted above.

2. A 3-5 minutes video demonstration of your system while in operation. (Please explain the operation with your voice during recording and make sure to do rigorous testing to showcase your system's full operation – all group members are to take part in this)

3. All your C++ codes showing your detailed implementation with comments and compliance to good coding practice.